When July Gets Hot: Heat, Drought and U.S. Corn Yields

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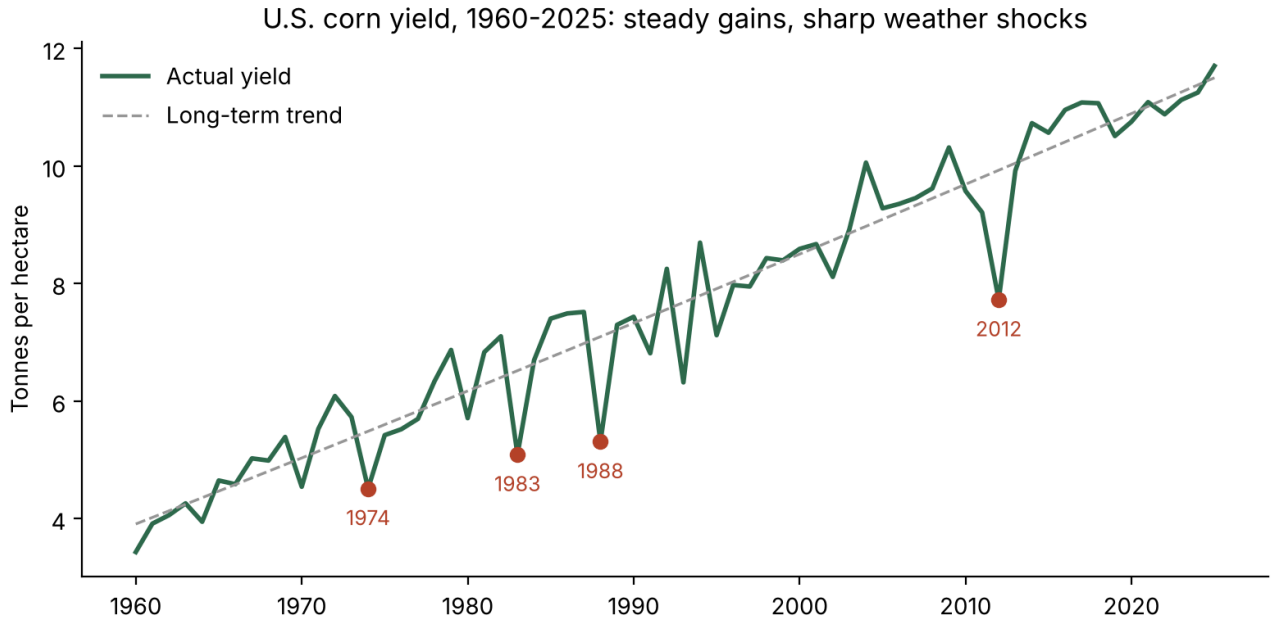
Summary. U.S. corn yields have more than tripled since 1960, but the worst years in that record line up with hot, dry Julys in the Corn Belt. Using 66 years of USDA/FAO yield data and NOAA weather records for Iowa and Illinois, I found that each 1°F of extra July heat is linked to about a **1.7% lower yield** than expected. The ten hottest and driest Julys averaged yields **10.9% below trend**, while all other years came in 1.9% above it. As summers warm, protecting yields from heat and drought is one of the most direct ways research can keep production systems strong.

Why this question matters

Corn is the largest crop in the United States and feeds into livestock feed, food products and fuel. When yields drop sharply, as they did during the 2012 drought, the effects spread to farm incomes and food prices. July matters most because that is when corn pollinates, and heat and dry soil during pollination can permanently reduce the number of kernels on each ear.

Data and method

- **Yields:** U.S. corn yield in tonnes per hectare, 1960–2025 (FAO and USDA data, via Our World in Data).
- **Weather:** average July temperature and total July rainfall for Iowa and Illinois, the two largest corn-producing states (NOAA National Centers for Environmental Information).
- **Removing the technology trend:** yields rise almost every decade because of better seeds and farming practices, so I fit a long-term trend line and measured each year as a percent above or below it. This separates weather shocks from long-run progress.
- **Model:** a simple linear regression of each year's yield gap on its July temperature and rainfall. All analysis was done in Python (pandas, NumPy, Matplotlib).

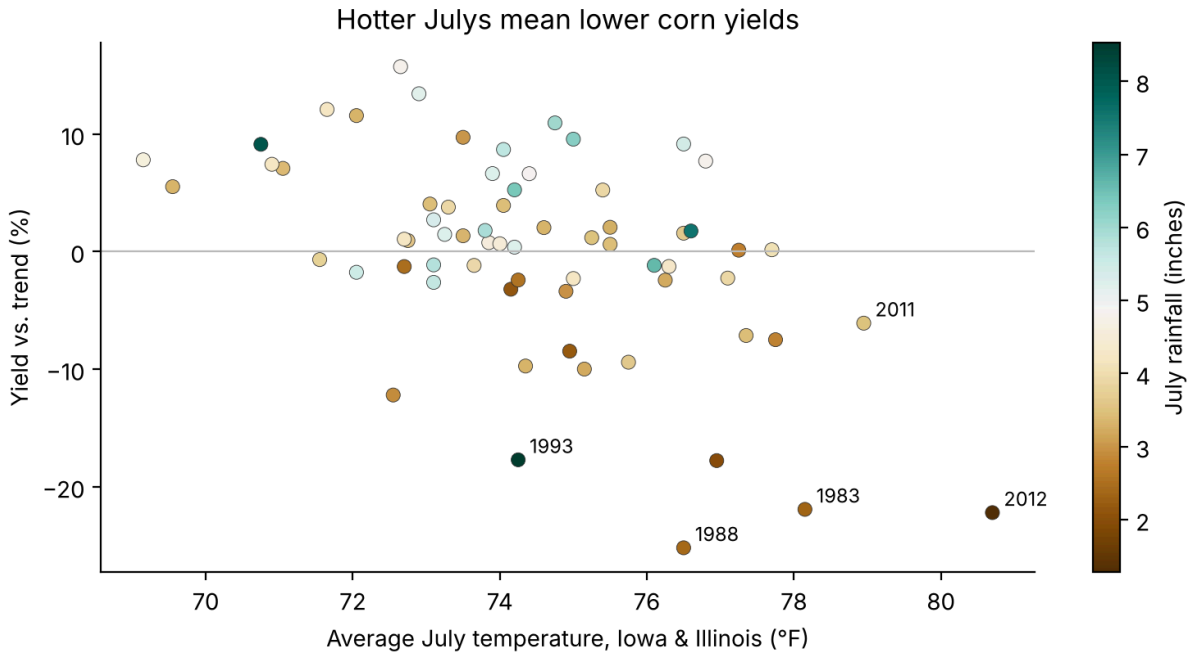


Sources: FAO/USDA via Our World in Data (yield); NOAA NCEI Climate at a Glance (July weather, Iowa & Illinois).

Figure 1. Yields climbed from 3.4 to 11.7 tonnes per hectare, but four years fell more than 17% below trend.

Findings

1. Heat is the clearest warning sign. July temperature had a stronger link to yield gaps (correlation −0.52) than July rainfall (+0.39). In the regression, each 1°F above the average July temperature (74.4°F) came with about 1.7% lower yield, and each extra inch of rain with about 1.6% higher yield. Together, July heat and rain explain about a third of the year-to-year swings, which is a lot for just one month of weather.



Sources: FAO/USDA via Our World in Data (yield); NOAA NCEI Climate at a Glance (July weather, Iowa & Illinois).

Figure 2. Each dot is one year. The worst years (bottom right) were hot *and* dry; dark green dots are very wet Julys.

2. The biggest losses came when heat and drought hit together. The four worst years in the record were all hot, dry summers. The fifth, 1993, was the opposite:

Year	Yield vs. trend	July temp (°F)	July rain (in)	What happened
1988	−25.2%	76.5	2.4	Severe Midwest drought
2012	−22.2%	80.7	1.3	Hottest, driest July in the record
1983	−21.9%	78.2	2.3	Summer heat wave and drought
1974	−17.8%	77.0	2.0	Hot, dry summer
1993	−17.7%	74.3	8.5	Great Flood of 1993: too much water

In 2012 alone, yields came in about 2.2 tonnes per hectare below where the trend said they should be.

3. Too much water hurts too. 1993 is the exception that proves the point: July was not hot, but record flooding cut yields almost as much as the drought years. The relationship with rain is not simply "more is better." Crops do best in a middle range, which a straight-line model cannot fully capture.

What this means for research

Better seeds and practices have raised yields for decades, but they have not removed the weather risk. The 2012 drought cut yields almost as much, in percentage terms, as the 1988 drought did. This connects to two of FFAR's priority areas:

- **Cultivating Thriving Production Systems:** crop varieties that tolerate heat during pollination, and planting choices that move the most sensitive growth stage away from peak heat, target exactly the losses shown here.
- **Sustaining Vibrant Agroecosystems:** healthier soils hold more water, which can soften a dry July. Better drainage and water management help in flood years like 1993.

Since warmer summers are expected to become more common, measuring how much of this risk new practices actually remove would be a useful next step.

Limitations

This is a simple model built for clarity. It uses national yields with weather from two states, one month of weather, and a straight-line effect for rain, so it shows strong associations rather than proof of cause. A fuller analysis would use county-level yields from USDA NASS, daily temperatures (especially hours above 86°F), and soil moisture data.

Sources: Our World in Data, "Corn yields" (FAO and USDA data), ourworldindata.org/grapher/maize-yields; NOAA National Centers for Environmental Information, Climate at a Glance: Statewide Time Series, Iowa and Illinois, July average temperature and precipitation, ncei.noaa.gov. Code and data: available on request.